# Choosing Our Solution

**Score every idea the same way, then pick the best one together.**

**Name:** _____ **Date:** _____

**Our problem**

Invasive grasses (Guinea, Gamba, Molasses and Cogon) dry out and burn. The fires kill rainforest plants, and the grass grows back first, so every fire lets the grass spread further into the rainforest. Our solution has to help break this **grass–fire cycle**.

## What to do

1. **Read** the scoring guide below so we all score the same way.
2. **Score each idea on your own** on page 2. Give each idea 1 to 5 points in each column. Use the research, not just what sounds cool.
3. **Got a new idea?** Write it on page 3 and score it the same way.
4. **Add up as a team** on page 4. Then talk about the top two before we vote.
5. **Ask an expert.** Write what they tell us on pages 5 and 6, and what we changed because of it.

## Scoring guide

| Score it on … | 1 No | 3 Some | 5 Yes! |
|---|---|---|---|
| **New**<br>Is it innovative? | Someone already does this, the same way. | Uses something that exists, but in a new way or a new place. | Nobody does this yet. We can say what is new. |
| **Doable**<br>Could it really work? | Too big, too costly or too risky. We can't explain how it works. | Could work with more research. We can draw it. | Could really work in a rainforest, and we can show it with a model by 11-17. |
| **Biodiversity**<br>Does it help plants and animals? | Doesn't help, or harms native plants, animals or soil. | Helps one part: stops the grass **or** stops the fires. | Breaks the whole cycle and protects lots of rainforest life. |

Use 2 and 4 for in between. The highest total is 15.

**Judges will ask**

- What solutions already exist, and how is ours different?
- How does your solution help biodiversity?
- Who gave you feedback, and what did you change because of it?

# My scores

**Name:** _____

Give 1 to 5 points in each box. Add the three boxes for the total (out of 15). The grey words under each idea are the big question we still need to answer.

| Idea | New | Doable | Bio-diversity | Total /15 |
|---|---|---|---|---|
| **A. Fire drones: find fires and put them out**<br>Do rainforest fire drones already exist? What would we do that is new? | | | | |
| **B. Drones that drop sand on fires**<br>Does sand put out grass fires? Is it safe for the soil? | | | | |
| **C. Map and watch the grass from above**<br>Can drones or satellites spot the grass early, and how accurately? | | | | |
| **D. Winged drones that find and fight fires**<br>Can a winged drone carry and drop fire-fighting material? How does it take off and land? | | | | |
| **E. Herbicide beside roads**<br>Harms native plants? Is it new at all? | | | | |
| **F. Map the grass, then send a drone to the spots about to burn**<br>Can we join C and D into one system? | | | | |
| **G. New idea (page 3):**<br>_____ | | | | |
| **H. New idea (page 3):**<br>_____ | | | | |

✎ **My favourite is _____ because ...** (use a fact from our research)

✎ **One thing that worries me about it is ...**

## + Our new idea

DESIGN

**TIP**
A new idea can be brand new, or two ideas joined together, or an old idea changed to work better. Score it with the same guide as the others.

**Idea name:** _____

**Who thought of it:** _____

**What it does, and how it works:**

**Which part of the cycle does it stop?**

☐ the grass spreading ☐ the fires starting ☐ the fires spreading ☐ the grass growing back

**Does something like this already exist? What is new about ours?**

**The big question we need to answer, and who could help us:**

**Quick sketch**

Draw it. Label the parts.

| New | Doable | Biodiversity | Total /15 |
|---|---|---|---|
| | | | |

## Team scores

Each person reads out their total for each idea (from page 2). Add across for the team total, out of 135.

| Idea | Sylvie | Brynn | Jack | Fedor | Ira | Jordan | Ella | Alexis | Amelia | Team total |
|---|---|---|---|---|---|---|---|---|---|---|
| **A. Fire drones** | | | | | | | | | | |
| **B. Sand drones** | | | | | | | | | | |
| **C. Map the grass** | | | | | | | | | | |
| **D. Winged drones** | | | | | | | | | | |
| **E. Herbicide by roads** | | | | | | | | | | |
| **F. Map, then send a drone** | | | | | | | | | | |
| **G.** | | | | | | | | | | |
| **H.** | | | | | | | | | | |

## ✓ Our choice

DESIGN

**Top two ideas:** _____ **and** _____

**We chose:** _____ **Votes:** _____

✎ **We chose it because ...** (new? doable? helps biodiversity? give a fact)

✎ **It beat _____ because ...**

✎ **What we still need to prove, and who will find it:**

✎ **On 11-17 we will build or draw:** (prototype or detailed drawing)

# Expert feedback

Fill in a box each time an expert replies: by email, on a call or in person. Write what they really said. This is how we show judges what we changed and why. Experts we have asked or may ask: Bob Leigh (RTI), Matthew Lusty (GE Aerospace), the ROM, Toronto Botanical Garden, Allen Gardens, and Brynn's firefighter.

**Expert:** _____ **Works at:** _____

**Date:** _____ **How:** ☐ email ☐ call ☐ visit **Written down by:** _____

✎ **What we asked:**

✎ **What they told us:**

✎ **What we will change because of it:** (or why we kept it the same)

**Expert:** _____ **Works at:** _____

**Date:** _____ **How:** ☐ email ☐ call ☐ visit **Written down by:** _____

✎ **What we asked:**

✎ **What they told us:**

✎ **What we will change because of it:** (or why we kept it the same)

# More expert feedback

Feedback from a coach, a parent or another team counts too. Record who said it and the date.

**Expert:** _____ **Works at:** _____

**Date:** _____ **How:** ☐ email ☐ call ☐ visit **Written down by:** _____

✎ **What we asked:**

✎ **What they told us:**

✎ **What we will change because of it:** (or why we kept it the same)

**Expert:** _____ **Works at:** _____

**Date:** _____ **How:** ☐ email ☐ call ☐ visit **Written down by:** _____

✎ **What we asked:**

✎ **What they told us:**

✎ **What we will change because of it:** (or why we kept it the same)